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THE ORACLE THAT WEARS US

Project:

Interactive Installation

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20220711**Field:**

Fashion & Emerging Tech

The Oracle That Wears Us is an arcade-style installation housed within a projection-mapped tent. Authorship shifts from designer to machine to public as inputs are captured, processed through AI engines, and materialized via a robotic pen plotter onto fabric fragments.

Terms and Abbreviations

Term	Definition
Big M	Major stage marked as GitHub Milestones (e.g., Research).
OSC	Open Sound Control — real-time communication protocol.
Typst	Modern typesetting system used for automatic beautiful reports.
MVP	Minimum Viable Product — core working version of the installation.
PR	Pull Request — code review step before merging.
Tiny Sprint	Small daily or 2-day task (1–4 issues per person).
Project Board	GitHub Kanban board for task tracking (Backlog → Done).

How the Project Timeline Works

The timeline uses a “Main + Daily Sub-lines” structure to guarantee visibility:

- **Main timeline:** Protected `main` branch (only stable, reviewed code).
- **Weekly phases:** Temporary `week-xx` branches for one-week goals.
- **Student sub-lines:** Personal branches starting with `name/`.

Weekly Flow:

1. Monday: Team meeting, issues moved to “This Week”.
2. During week: Work in individual branches with tiny PRs.
3. Sunday night: GitHub Action generates this PDF report.
4. Monday morning: Report + live demo to the professor.

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CORE CONCEPT & STRUCTURE

Tech Summary

- **Hardware:** Plotter, arcade console, projector, camera.
- **Software:** Unity, TouchDesigner, OSC, Python, Generative AI.

Team Roles (High-Level Responsibility)

- **Nadine Allan:** System Architecture & Integration.
- **Andrey Dyakov:** Plotter & Materialization.
- **Dmitri Kazantsev:** Generative Concepts & Fabrication.

Agile Setup

- **Big Milestones:** 6 Phases (GitHub Milestones).
- **Small Milestones:** One per week (due Sunday).
- **Tiny Sprints:** Daily/2-day GitHub Issues.

Important Notes & Flexibility:

The project can (and will) morph based on research and testing. Possible changes include single cumulative fabric vs modular pieces or UV-hidden layers. All decisions will be documented in weekly reports.

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PROJECT TIMELINE

Phase 1 – Research

Task	Deliverable
Research core problems: algorithmic mediation & collective creation	Problem statement + literature summary
Review similar interactive installations	Comparison table & lessons learned
Hardware/Software evaluation	Final tech stack (Plotter, OSC, AI)

Phase 2 – Experimentation

Task	Deliverable
Test interaction methods (gestures/buttons)	Working prototypes (2-3 approaches)
Generative techniques & visual styles	Pattern output comparison
Plotter control & fabric tests	Selection report + test drawings

Phase 3 – System Architecture

Task	Deliverable
Finalize OSC schema & signal chain	Locked architecture document
Unity arcade prototype with OSC	Working input & timing capture
OSC Layer (Unity TD Python)	Stable real-time data flow

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FABRICATION & FINALIZATION

Phase 5 – Fabrication & Spatial Setup

Task	Deliverable
Projection mapping on tent surface	Calibrated projection
Physical arcade machine build	Finished console
Safety & risk plan	Complete safety document

Phase 6 – Testing & Polish

Task	Deliverable
Full system rehearsals	Stable end-to-end installation
Bug fixing & latency optimization	Tuned and reliable system
Final documentation & video	Complete Delivery 2 materials

MVP and Extra Ideas Pool:

We maintain a clear MVP edition (arcade → patterns → fragments). Extra ideas (UV-hidden layers, degeneration logic) will be activated only after MVP sign-off and ahead-of-schedule progress.